AI-Powered Intelligent Job Search and Career System

Data Science and AI Project

Milestone 5 Report

Group 4

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Index

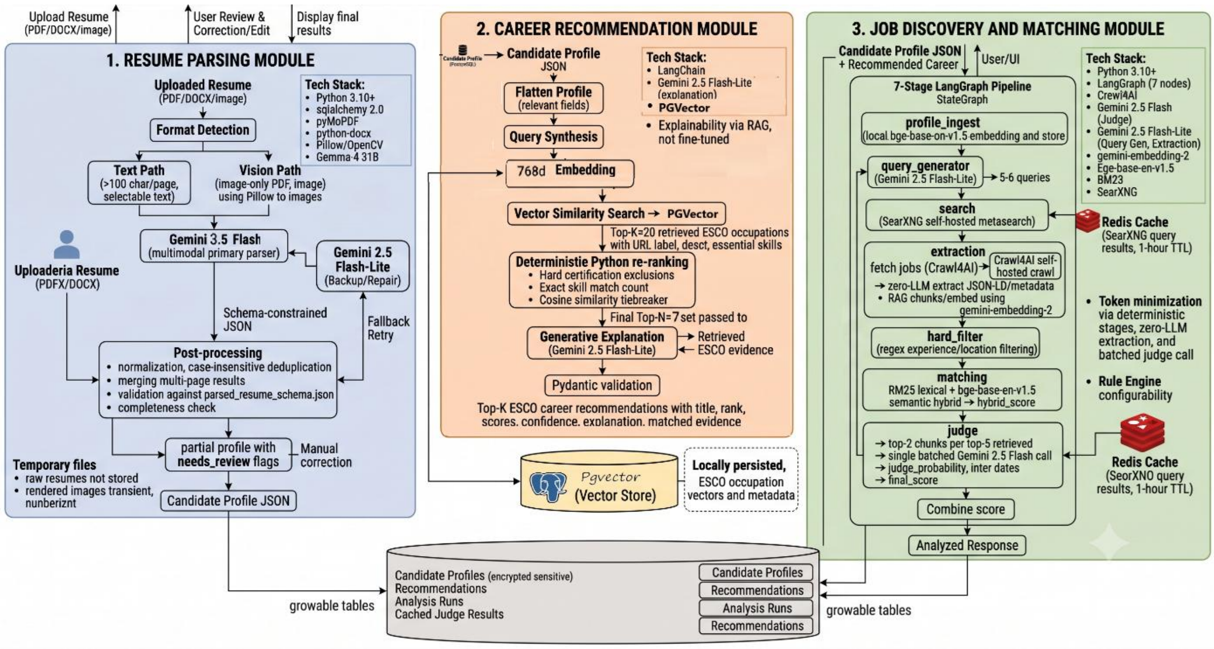
1. Introduction

1.1 Project objective and scope of this evaluation

The AI-Powered Intelligent Job Search and Career System converts an uploaded resume into a structured candidate profile, maps it onto occupations in a standard taxonomy, and uses that role context to discover and rank live job postings. The three modules — Resume Parsing, Career Recommendation, and Job Discovery and Matching — communicate through a shared candidate profile and a shared database.

Milestone 5 is the evaluation and integration milestone. All three modules now run inside one FastAPI application over one centralised Supabase database with the vector store on pgvector. Resume Parsing is evaluated field by field over 86 annotated resumes; Career Recommendation is evaluated on the same set with retrieval and ranking metrics on both the full set and a held-out split; and Job Matching, Job Discovery and the two prompt-driven stages are evaluated through a deterministic harness across configuration sweeps. Results are reported with confidence intervals, and components that do not meet their acceptance criteria are reported as failing.

1.2 Model Architecture



The models used at each inference stage are:

Module	Component	Model
Resume Parsing	Primary parser	Gemini 3.5 Flash, vision-only at 150 DPI
Resume Parsing	Evaluated baseline	Gemma 4 31B (multimodal, open weight)
Resume Parsing	Backup and repair	Gemini 2.5 Flash-Lite
Career Recommendation	Embedding model	BAAI/bge-base-en-v1.5, 768 dim, local CPU
Career Recommendation	Vector store	Supabase PostgreSQL, pgvector, HNSW cosine index
Career Recommendation	Explanation model	Gemini 3.5 Flash-Lite
Job Discovery and Matching	Query generation	Gemini 2.5 Flash-Lite
Job Discovery and Matching	Judge	Gemma 4 31B
Job Discovery and Matching	Embedding model	BAAI/bge-base-en-v1.5, 768 dim, local CPU

1.3 Nature of the work in this milestone

Considering GenAi, this system performs no gradient-based training. No weights are initialised, updated, or saved anywhere in the pipeline, because the project has no labelled training data of the kind these tasks would require. Conventional headings such as optimiser, learning rate, epochs, and checkpoint selection have no direct instance. They are mapped to their configuration-tuning

equivalents rather than skipped. Every improvement reported here is a change of prompt, schema, retrieval parameter, scoring weight, or provider routing.
Our training has been replaced with this.

Conventional activity	Equivalent in this system
Trainable weights, optimiser, learning rate	None; configurable prompts, schemas, retrieval parameters, and scoring weights chosen by exhaustive sweeps over small discrete grids
Batch size	A cost measure only: the judge scores five shortlisted jobs per call; the parser sends one page image per call
Number of epochs	No repeated passes over data; the occupation index is built in a single pass
Loss function	Deterministic rule-based validators combined into a weighted composite score, maximised by each sweep
Early stopping	A three-condition stopping criterion for prompt refinement; a fixed exhaustive grid for parameters
Checkpoint selection	Configuration-bundle selection; the winning bundle is the artefact the system runs on
Regularisation	Schema constraints, evidence grounding, URI validation, & permissive filtering
Training duration, loss curves	Index build time and evaluation runtime; the sweep response curve is the equivalent diagnostic

2. Experimental Setup

2.1 Environment, frameworks, and libraries

Layer	Detail
Compute	No GPU anywhere. Gemma 4 31B and the Gemini-family models run over hosted APIs (Vertex AI for the parser); BAAI/bge-base-en-v1.5 (110M params) is the only local model, on CPU
Language and application	Python 3.10+ managed with uv; one FastAPI application exposing all module routers; SQLAlchemy 2.0 async with Alembic migrations
Storage	Supabase PostgreSQL (ap-northeast-2) with pgvector; HNSW index using the cosine operator class; psycopg2 for ingestion writes
Orchestration and ranking	LangChain in Career Recommendation; a seven-node LangGraph StateGraph in Job Discovery; rank_bm25 for lexical scoring; sentence-transformers for embedding
Discovery services	Azudhan job API as the primary source; SearXNG and Crawl4AI self-hosted as fallback, with a one-hour Redis result cache
Evaluation harness	promptfoo-based suite with rule-based Python validators; append-only JSONL evidence ledger recording outputs, schema evidence, errors, latency and tokens; pytest for the unit and contract test suite; a manual end-to-end record of 20 test cases across the integrated path, including negative cases

All grading is deterministic and rule-based rather than model-graded, so the evaluation itself stays reproducible. Parser scoring reuses saved predictions and never makes an additional model call; gold and normalisation hashes prevent stale scores being reused.

2.2 Datasets, versions, and splits

Dataset	Version and size	Role and split
Resume corpus	86 records, 58 source-verified	Gold set, 2 per category across 43 categories; dev 34, held-out 52
Occupation taxonomy	ESCO v1.2.1: 3,039 occupations	Retrieval knowledge base, indexed in full, not split
Career Recommendation labels	134 category-to-occupation mappings	Ground truth for retrieval metrics; inherits the resume split
Job Matching golden set	86 candidates × 10-job pool	Ranking evaluation with real BM25 and embeddings; 60/20/20 candidate split

Job Discovery live queries	Multiple domains and locations exercised against the API path, including explicit failure cases	Live search reliability; still approximately one query per domain
Prompt / judge evaluation	4 profiles sourced from the tested recommendation stage; job evaluation dataset used for batches, now including negative cases	Prompt sweeps executed as real calls

This broadens domain coverage and adds negative cases, which the earlier evaluation lacked. It remains a moderate rather than large-scale evaluation: the domain count improved, but per-domain sample depth on the live-search side is still shallow, and that limit is carried into the limitations rather than discharged.

2.3 Reproducibility

- Random seed 42 fixed across split assignment and sampling; temperature 0 on every generative call.
- Schema-constrained output on every generative call, validated before downstream use; complete model and prompt fingerprints recorded per run.
- Model identifiers held in environment configuration, so a retired model can be swapped without a code change.
- Database schema built from versioned Alembic revisions and a checked-in setup script; gold corrections stored as versioned overlays that preserve the original annotations.

2.4 Evaluation workflow

Each component follows the same loop: the configuration surface and metrics are fixed before the sweep runs, alternative configurations are run with everything else held constant, each is scored by a deterministic validator into one weighted composite, the highest composite is selected automatically, and the winner is re-run against an explicit baseline.

3. System Configuration Summary

3.1 Integrated architecture

The system run on shared FastAPI application, reading the persisted recommendation run by profile identifier. The full request path is:

Step	Endpoint	Behaviour
1	Resume upload	Vision parser returns a validated profile; sensitive fields encrypted at rest; profile identifier streamed back over the event stream
2	Recommendation	Reads and decrypts the stored profile, translates schema across the module boundary, embeds, retrieves top-20, re-ranks, calls the explanation model, persists a run record with provenance
3	Job search	Seeds target roles from the stored recommendation run; Azudhan API primary, SearXNG and Crawl4AI fallback; hybrid BM25 and embedding ranking; batched judge
4	Career report	Orchestrates the above sequentially and generates the report and PDF

3.3 Final configuration bundle

Each value is labelled empirical (selected by a measured sweep) or heuristic (set by reasoning, not yet varied and measured).

Module	Parameter	Value	Basis
All	Temperature	0	By design — reproducibility required
All	Embedding dimensions	768	Fixed by the encoder

Resume Parsing	Provider and route	Gemini 3.5 Flash vision-only, 150 DPI, medium thinking	Empirical — beat Gemma 4 on every section F1
Resume Parsing	Skill boundary rule	Preserve competency phrases; split explicit enumerations	Empirical — six prompt experiments, Skills F1 0.2268 to 0.8198
Resume Parsing	Description rule	Copy visible text without summarising	Empirical — exact copy rate and description similarity measured
Career Rec.	Chunk contents	Preferred/alt labels, description, essential & optional skills	Empirical in part — optional skills present for 2,857 of 3,039 occupations
Career Rec.	Vector distance	Cosine, explicit HNSW operator class	Empirical — measured against default-distance index
Career Rec.	Skill-bonus weight	0.02	Empirical — eight values swept against MRR and Hit Rate
Career Rec.	Essential/optional skill weights	1.0 and 0.5	Heuristic — ratio not swept
Career Rec.	Retrieval Top-K and Top-N	20 and 7	Empirical in part — Hit@20 of 0.9186 confirms the retrieval window is wide enough; Top-N compared at 5 and 7, with 7 selected (Hit@7 0.8488 against Hit@5 0.8372).
Job Matching	BM25 / embedding weights	0.5 and 0.5	Empirical — six configurations compared
Job Matching	Shortlist, hybrid, judge weights	15; 0.8 and 0.2	Heuristic — not varied in the sweep
Job Discovery	Source routing	Azudhan API primary; SearXNG and Crawl4AI fallback	Empirical — API path validated across domains and locations
Job Discovery	Search and crawl parameters	max_results 5, crawl limit 3, concurrency 3, timeout 8,000 ms	Empirical — five parameter sets compared

4. Evaluation Methodology

4.1 Protocol

Each component is evaluated offline against a fixed reference set, isolated from the stages around it. Resume Parsing is scored field by field against the corrected gold set. Career Recommendation is scored on retrieval and re-ranking only, since the explanation call cannot change the measured ranking. Job Matching is scored on a fixed candidate-job pool to separate ranking quality from crawl reliability. Job Discovery is scored on live queries because search reliability is what is being measured.

4.2 Ground truth

58 of the 86 gold records have been manually verified against their source PDFs, with corrections stored as versioned overlays so the original annotations remain auditable. The corrections were not cosmetic; they changed what the evaluation measures.

Field	Correction applied	Representative example
Skills	Replaced empty or inferred lists with entries visibly printed in skill-like sections; preserved competency phrases; removed duplicates	A DevOps record had empty gold skills despite visible Kibana, Prometheus, Datadog, CI/CD, Jenkins, Terraform, AWS, Azure, GCP, Docker and Kubernetes
Education	Corrected degree/field boundaries, nested credentials, institutions and date direction	“Advanced Technical Certificate” restored to the complete visible “Advanced Technical Certificate in Automotive Technology”

Experience descriptions	Replaced abbreviated AI summaries with text grounded in the corresponding source entry	A summary reading "expertise in data warehousing" replaced with the complete ordered duties as printed
Projects	Added visibly named patents, publications and projects; removed awards and generic duties incorrectly labelled as projects	A visible conference paper added as a project, with the presentation sentence retained as its description
Certifications	Added visible credentials missing from empty gold lists; separated credentials from awards, affiliations and degrees	A Data Science record corrected to include four visibly printed credentials
Record identity	Corrected mismatched source-to-profile associations found during side-by-side review	One web-design record was linked to the wrong reference profile and was replaced with the profile belonging to its source PDF

The Career Recommendation ground truth is unchanged: each of the 43 resume categories maps to a set of acceptable occupations (134 mappings), generated by keyword matching against taxonomy titles and reviewed by hand. It remains a proxy and supports rank-position measures rather than set-coverage measures.

4.3 Cross-validation strategy

No k-fold cross-validation is used: the gold set holds two resumes per category, so a fold would place a single resume per category, producing unstable per-fold estimates. A single stratified development and held-out split is used instead, with near-duplicate resume groups confined to one side so a visually similar resume cannot straddle the boundary.

4.4 Baselines

Component	Baseline	Why this baseline
Resume Parsing	Gemma 4 31B on the same 86 resumes and the same corrected gold set	Isolates the provider and prompt change from the scoring change
Career Recommendation re-ranking	Retrieval alone at Top-K 20 (pass-through); previously specified exact-skill-match rule	The floor the re-ranker must beat, and the design it replaced
Career Recommendation retrieval	Pre-migration index using the default distance function	Isolates the effect of the vector-store migration
Job Matching ranking	bm25_only, embedding_only, previously configured 0.4/0.3 default	Tests whether the hybrid score beats either signal alone
Job Discovery and the prompt stages	The previously configured default parameter set. the baseline prompt variant	Tests whether sweeping and prompt engineering contribute anything beyond what was already in place

4.5 Acceptance criteria and outcomes

Criterion	Outcome
Resume Parsing: per-field F1 ≥ 0.75 across all sections	Met — all six sections pass; lowest is Certifications at 0.75 (0.7472 unrounded)
Resume Parsing: no schema violation on the development split	Met — after the Vertex structured-response correction, schema validity moved from 0/5 to 5/5 on the smoke cohort
Career Rec.: re-ranking not below retrieval on primary measures	Met on every primary measure: Hit@1 rises from 0.4651 to 0.4884, Hit@5 from 0.8256 to 0.8372, and MRR from 0.6110 to 0.6121
Career Rec.: correct occupation within the initial retrieval window	Met — Hit@20 is 0.9186
Career Rec.: zero unrecoverable failures	Met — 86 of 86 profiles produced a ranked list

Job Matching: ranking quality ≥ default weights	Met — matches or exceeds the default composite and is an order of magnitude faster
Job Discovery: search coverage sufficient	Met — coverage 1.0 in every configuration; the API path returns results across multiple domains and locations within two minutes
Job Discovery: crawl success reliable	Not met — crawl remains 0.50; the re-architecture reduces how often that layer is reached but does not repair it
Judge stage: schema-valid, latency-bounded, calibrated output	Not met — no prompt variant passes; root cause identified as token overflow
Integration: all modules over one application and database	Met — Job Discovery now consumes the stored recommendation run inside the shared application
Human evaluation of recommendation usefulness	Not evaluated — no human study was run; protocol specified in Section 4.6

4.6 Human evaluation: status and specified protocol

The evaluation in this report is entirely automated. No user satisfaction, explanation-quality, or usefulness study was conducted, and reporting an estimate would be worse than reporting the gap. The gap is stated, and the protocol that would close it is specified below.

Dimension	Proposed instrument	Why it is not substitutable by an automated metric
Recommendation usefulness	Candidates rate each of the top 7 occupations as plausible, adjacent, or irrelevant to their own career intent	The taxonomy proxy measures whether an acceptable occupation is retrieved, not whether the candidate would act on it
Explanation quality	Rated for factual support against the retrieved evidence and for informativeness, on a fixed scale, by at least two raters	URI validation proves the explanation cites only retrieved occupations; it says nothing about whether the reasoning is useful
Job match relevance	Recruiters or domain reviewers judge the top 5 postings per profile	The judge stage that would automate this is the component currently failing
Inter-rater agreement	Cohen's kappa across raters before any score is reported	A single unvalidated rater would be no more trustworthy than the automated proxy it replaces

A study of roughly 20 to 30 profiles with two raters would be sufficient to detect a coarse quality signal at this system's scale and is stated as required future work rather than as a commitment already met.

5. Performance Metrics

The three modules pose different measurement problems. Resume Parsing is an extraction task and takes field-level precision, recall and F1. Career Recommendation and Job Matching are ranking tasks and take retrieval and rank-correlation metrics. Job Discovery is a reliability problem and takes success rates. The prompt-driven stages are measured on structural validity, internal consistency, and latency.

Metric	Definition and why it is used
Precision, Recall, F1	TP/(TP+FP), TP/(TP+FN), 2TP/(2TP+FP+FN), micro-aggregated per section so sparse fields cannot dominate. Item-level true negatives are not defined, because the set of possible skills, employers and degrees is unbounded
Description similarity	Coverage plus TF-IDF cosine, used only for long free-text fields where faithful copy can differ character by character
Hit Rate at K	Fraction of profiles with an acceptable occupation in the top K — what the user actually sees
MRR	Mean reciprocal rank of the first acceptable occupation; the primary ordering measure

Precision at 5, NDCG at 10, Spearman	Job Matching ranking quality; Spearman is the most sensitive on a ten-item pool
Wilson score interval	95% interval on a binomial proportion, used because the normal approximation is unreliable at these rates and sample size
Coverage and crawl success	The two places the discovery pipeline can fail; crawl success is in practice the binding one
Schema, diversity, calibration scores	Structural validity of prompt-stage output, query duplication, and whether a categorical judgment agrees with its own probability

Two metrics used earlier in the project were dropped: Recall at 100, because the system never shows a candidate 100 occupations, and NDCG at 20 with Precision at 20, replaced by the at-10 and at-5 forms for the same reason.

5.1 Metrics deliberately not used

- **BLEU/ROUGE/BERTScore:** no stage generates free text against a reference; the explanation model is checked by URI validation against retrieved evidence instead.
- **ROC-AUC / PR-AUC:** the match score is a deterministic decision-support value, not a calibrated probability.
- **Confusion matrix:** no small, fixed label set; ranking runs over 3,039 occupations. The informative equivalents are the per-field confusion examples in Section 6.1 and the quantified miss decomposition in Section 10.1.
- **Hallucination rate:** prevented structurally by URI validation rather than measured statistically; the guard never needed to fire across the annotated runs.

6. Experimental Results

6.1 Resume Parsing

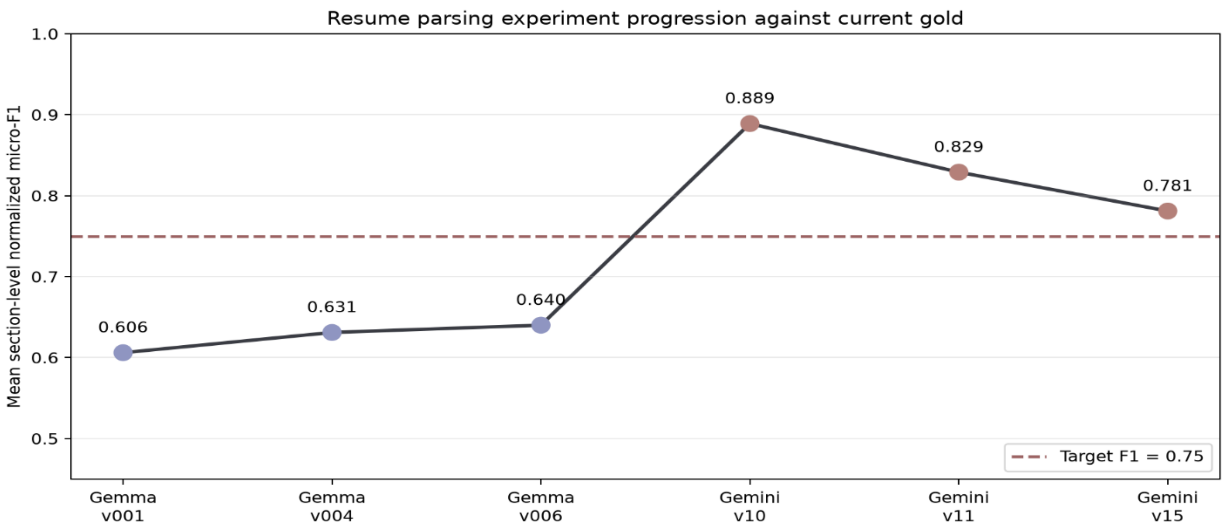


Figure 1. Experience description scoring across all 86 resumes. Coverage is near complete; the binding measure is text similarity, and both similarity figures clear the 0.75 acceptance target.

Resume Parsing was evaluated over all 86 gold resumes through the production vision route, scored field by field against the corrected gold set. All six sections meet the 0.75 acceptance threshold.

Section	TP	FP	FN	Precision	Recall	F1	F1 ≥ 0.75
Contact	139	3	2	0.98	0.99	0.98	Pass
Skills	1,190	395	128	0.75	0.90	0.82	Pass
Education	353	31	28	0.92	0.93	0.92	Pass
Experience	996	27	17	0.97	0.98	0.98	Pass
Projects	52	9	5	0.85	0.91	0.88	Pass

Certifications	133	73	17	0.65	0.89	0.75	Pass

Certifications pass on the unrounded value of 0.7472 and is the weakest section, driven by precision of 0.65: awards, honours and professional affiliations are returned as credentials. A stricter boundary instruction was tested and rejected — it reduced full-run Certifications F1 from 0.7472 to 0.6503 by suppressing valid credentials, so a locally correct rule was discarded on its net effect. Skills carries the largest absolute false-positive count for the same structural reason: grouped source phrases and individually annotated tools use different boundaries.

Experience descriptions are scored separately, because faithful copy of long text can differ character by character. Corresponding descriptions were present for 99.52 percent of matched entries, mean TF-IDF cosine where present was 0.81, and the overall description score with missing positions counted as zero was 0.80, above the 0.75 threshold while still penalising omissions.

6.2 Career Recommendation

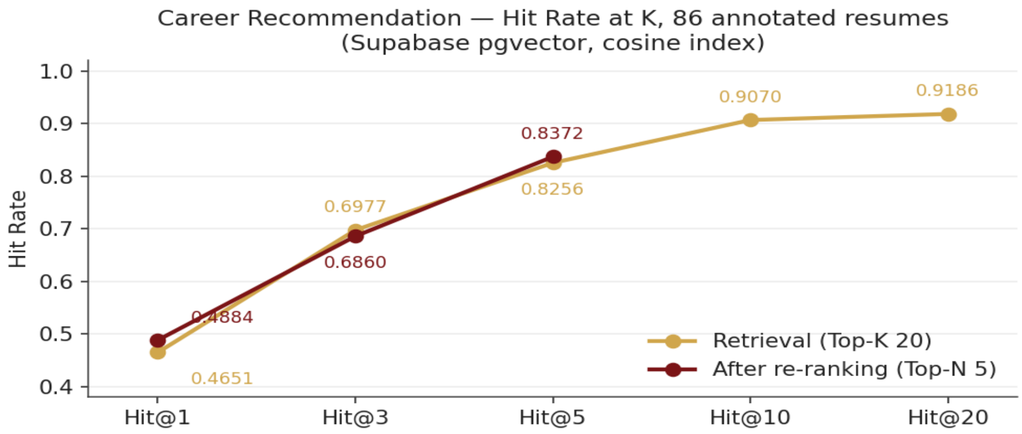


Figure 2. Hit Rate at K for retrieval and for the re-ranked output on the full annotated set. Re-ranking operates only within the top seven, so it has no value at K of 10 or 20.

Evaluated against the 86-record annotated gold set on the migrated index.

Stage	MRR	Hit@1	Hit@3	Hit@5	Hit@7	Hit@10
Retrieval, Top-K 20	0.6110	0.4651	0.6977	0.8256	—	0.9070
Re-ranked	0.6121	0.4884	0.6860	0.8372	0.8488	—
Held-out split, 52 resumes	0.6359	0.5192	0.7500	0.8269	—	—

The module produced a ranked list for every profile — 86 of 86 evaluated, 0 failures. On the held-out split the selected configuration matches retrieval at ranks 1 and 3 and improves Hit@5 from 0.8077 to 0.8269, sitting at or above the full-set figures, which is consistent with a configuration that is not overfitted to the set it was chosen on.

6.2.1 Statistical treatment

These rates are proportions over 86 records, so Wilson score intervals apply. The intervals are wide, and that width is itself a finding.

Measure	Count	Rate	95% Wilson interval	Width
Hit@1, retrieval	40 / 86	0.4651	0.363 – 0.570	0.206
Hit@1, re-ranked	42 / 86	0.4884	0.386 – 0.592	0.207
Hit@5, retrieval	71 / 86	0.8256	0.732 – 0.891	0.159
Hit@5, re-ranked	72 / 86	0.8372	0.745 – 0.900	0.155
Hit@7, re-ranked	73 / 86	0.8488	0.758 – 0.909	0.151
Hit@20, retrieval	79 / 86	0.9186	0.841 – 0.960	0.119

The headline Hit@5 difference is not significant. The improvement from 0.8256 to 0.8372 is one profile: 71 against 72 of 86. A paired exact test over the discordant pairs gives p = 1.00. The re-ranking step is therefore not shown to improve Hit@5. What the evidence supports is the weaker claim that it does not degrade the ordering, which the rule it replaced did.
The same caveat governs every configuration comparison in this report. At n = 86 the confidence interval on a rate near 0.83 is roughly plus or minus 0.08, so differences below about 0.08 are not separable. Sweeps are therefore read for the shape of their response curve rather than for the winning value, and single-run differences smaller than the interval width are described as not separable rather than as improvements. No repeated-trial variance is reported, because every generative call runs at temperature 0 and the deterministic stages are exactly reproducible; run-to-run variance is nil by construction, and the uncertainty that matters is sampling uncertainty over the 86 records, which is what the intervals above quantify.

6.3 Job Discovery and Matching

Four sweeps were run: two parameter sweeps on the stages that make no model call, and two prompt sweeps on the two stages that do.

6.3.1 Job Matching — ranking weight sweep, 86 candidates with 10 jobs each

Configuration	Composite	Precision@5	NDCG@10	Spearman	Latency (ms)	Pass rate
balanced_50_50 — selected	0.8977	0.8486	0.9795	0.8767	226	94.6%
embedding_heavy (0.3/0.7)	0.7670	0.8588	0.9793	0.8861	4,145	0%
bm25_only (1.0/0.0)	0.7656	0.8512	0.9793	0.8995	4,239	0%
current_default (0.4/0.3)	0.7593	0.8518	0.9791	0.8812	4,241	0%
embedding_only (0.0/1.0)	0.7518	0.8419	0.9706	0.8727	4,201	0%
bm25_heavy (0.7/0.3)	0.7503	0.8535	0.9789	0.8796	4,500	0%

Two limits are stated plainly. The API path does not return results for niche domains, which is exactly when the fallback is invoked and therefore when the 0.50 figure still governs. And the reported crawl-success number measures Crawl4AI in isolation, not the fallback chain end to end; re-running the Milestone 5 sweep methodology over the two-layer pipeline is required before a system-level reliability figure can be claimed. Crawl reliability is unchanged — the architecture reduces how often that layer is reached rather than making it more reliable.

6.3.3 Prompt sweeps and the judge stage

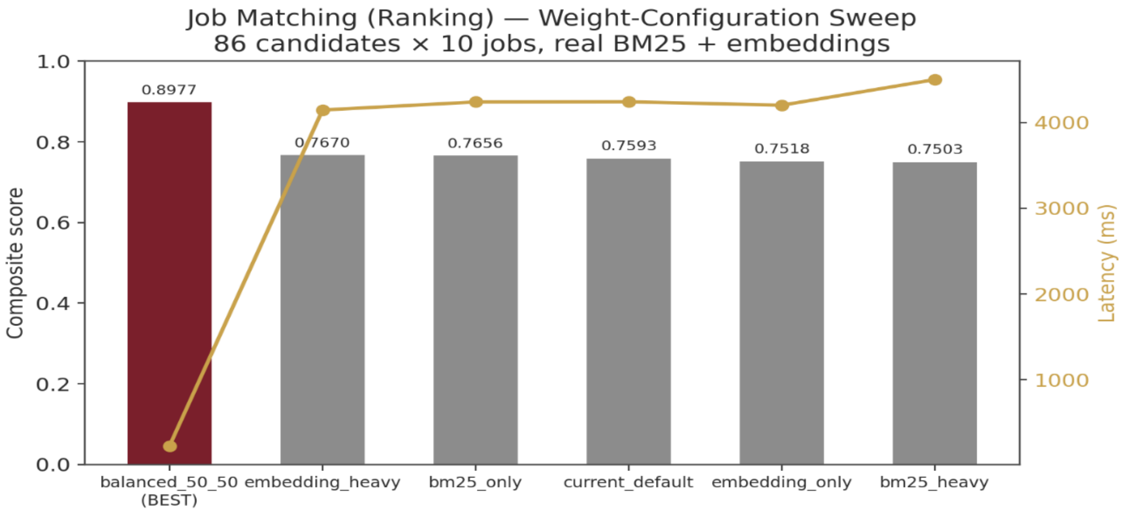


Figure 3. Composite score and latency across the six BM25/embedding weight configurations. Ranking-quality metrics are flat; latency decides the winner.

Stage and variant	Composite	Schema	Diversity / calibration	Latency (ms)	Pass rate
Query gen. — boolean_operators, selected	0.5102	1.0000	0.8010	19	100%
Query gen. — remaining six variants	0.3602–0.4306	1.0000	0.8010	26–33	100%
Judge — comparative, best available	0.2750	0.5000	0.5000	21,572	0%
Judge — narrative	0.2060	0.5000	0.5000	28,082	0%
Judge — baseline / weighted_criteria / strict_rubric / chain_of_thought	-0.0668 to -0.1500	0.0000	0.0000	27,872–35,716	0%

Root cause now identified. The judge failures across all six variants trace to token size rather than instruction wording. The combined payload — full resume text, full job description, instructions and any few-shot context — exceeds the usable context budget for the judge call, producing truncation-driven and hard failures. Six instruction styles at a 0 percent pass rate is consistent with a structural limit and not with a prompt-quality problem, which is why further rewording was abandoned.

A staged remedy is specified in order of effort: pre-flight token budgeting that counts tokens before the call and triggers a reduction step instead of sending and failing; structured extraction that passes skills, titles, years of experience, education and required qualifications rather than raw prose, which is the highest-leverage step because most of the token cost is text the judge does not need verbatim; and relevance-ordered truncation as a last resort where raw text must be retained. None of these is built, and the stage is reported as failing.

6.4 Integration verification

The merged path was exercised end to end on a real image-only resume from the corpus. The chain ran without manual intervention: schema-valid parse, profile identifier returned over the event stream, profile read and decrypted across the module boundary, embedded, retrieved, re-ranked, explained, and persisted, with job search seeded from the stored recommendation run.

Stage	Result
Parsing	Valid on the first attempt: fallback not invoked, no review flags, descriptions copied verbatim, personal identifiers correctly excluded
Normalisation	Six of seven skill bullets retained, the seventh correctly removed as a near-duplicate

Recommendation	Top 7: agricultural engineer, agricultural equipment design engineer, test engineer, agricultural machinery technician, agronomist, agricultural researcher, agricultural technician
Ranking accuracy	The candidate's actual occupation returned at rank 1, with all seven recommendations in the correct domain
Confidence handling	The resume writes skills as full sentences, so the deterministic overlap signal had nothing to match against short taxonomy labels. The system detected this, ranked on semantic evidence alone, returned the correct occupation first, and reported low confidence with an explicit statement of why
Job search	Target roles seeded from the stored recommendation run without manual input; the search, ranking and shortlist stages executed inside the same request. The batched judge ran but did not return usable output, consistent with Section 6.3.3

7. Baseline Comparison

7.1 Career Recommendation

Against the previously specified rule (exact skill-match count with similarity as tiebreak), the held-out improvement is large and separable: MRR rises from 0.4949 to 0.6359 and Hit@5 from 0.6731 to 0.8269. Against retrieval alone the change is small and mixed — Hit@5 improves by 0.0192 while MRR falls by 0.0078, both inside the interval width established in Section 6.2.1. The defensible claim is that the current step is a correct, inspectable, deterministic stage that does not degrade the retrieval ordering, which the earlier rule did.

7.2 Job Matching, Job Discovery, and the prompt stages

Component	Baseline	Selected	Improvement and nature
Resume parsing model	Gemma 4, aggregate micro-F1 0.6121 normalised	Gemini 3.5 Flash, all sections ≥ 0.75	Quality, separable in every section
Career Rec. re-ranking	MRR 0.4949 (previous rule)	MRR 0.6359	+0.1410, quality, separable
Career Rec. retrieval	Hit@5 0.7791 (default-distance index)	Hit@5 0.8256 (cosine index)	+0.0465, quality, not separable at n = 86
Job Matching ranking	Composite 0.7593 (configured default)	Composite 0.8977	+0.1384, efficiency only; ranking quality flat
Job Discovery search	Crawl 0.40 at 8,194 ms	Crawl 0.50 at 2,703 ms, plus an API primary layer	Both dimensions; crawl layer itself unimproved
Query generation prompt	Composite 0.3646	Composite 0.5102	+0.1456, efficiency; correctness already saturated
Batched judge prompt	Composite -0.0668, schema 0.0000	Composite 0.2750, schema 0.5000	Partial — structural validity only, stage still failing

7.3 Resume Parsing: Gemini 3.5 Flash against the Gemma 4 baseline

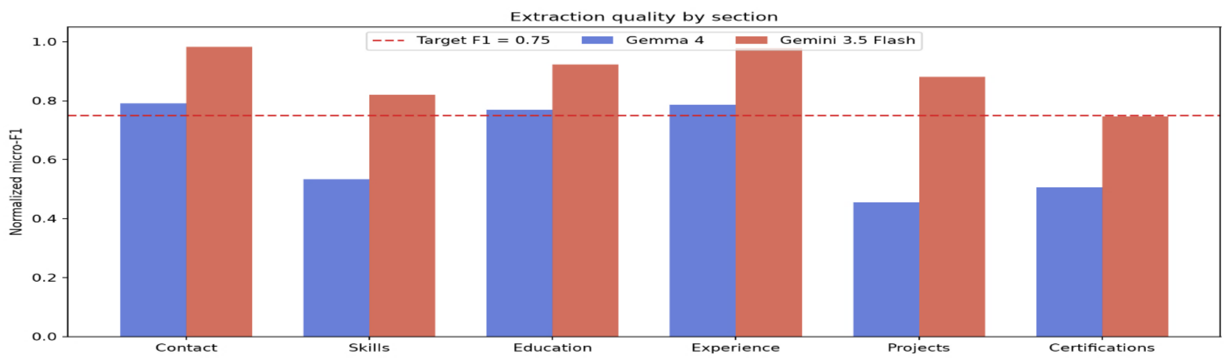


Figure 4. Section-level F1 for the selected parser against the Gemma 4 baseline on the same 86 resumes and the same corrected gold set.

Model	Contact	Skills	Education	Experience	Projects	Certifications
Gemma 4	0.79	0.53	0.77	0.79	0.45	0.51
Gemini 3.5 Flash	0.98	0.82	0.92	0.98	0.88	0.75

Gemini 3.5 Flash produced the higher F1 in every section, with the largest gains in the three weakest Gemma 4 sections — Projects +0.43, Skills +0.29, Certifications +0.24. Median latency was comparable (Gemma 4 at 17.21 s against 19.03 s), and Gemini 3.5 Flash had the lower worst case (80.26 s against 126.51 s); Gemma 4 also completed only 83 of 86 resumes against 86 of 86. The selection is justified by extraction quality, not by a claimed latency improvement. This compares the best completed configuration of each, so it measures the deployed model-and-prompt combinations rather than architecture alone.

8. Configuration Parameter Analysis

There are no hyperparameters in the training sense; the tunable surface is prompt text, output schemas, retrieval parameters, and scoring weights. Each sweep is an exhaustive grid over a small discrete set.

8.1 Career Recommendation — the skill-bonus weight

Skill-bonus weight	0.00	0.02	0.05	0.08	0.10	0.15	0.25	0.40
MRR	0.5849	0.5965	0.5866	0.5457	0.5328	0.4973	0.4866	0.4843
Hit@1	0.4651	0.4884	0.4767	0.4070	0.3953	0.3372	0.3256	0.3256
Hit@5	0.7791	0.7907	0.7907	0.7907	0.7674	0.7674	0.7674	0.7558

Weight 0.00 is a pass-through and exactly reproduces the retrieval ordering, confirming the implementation is correct and establishing the floor. Weight 0.02 improves on that floor on all three measures and was selected; above 0.05 every measure declines monotonically. The shape is the substantive result: the endpoint difference in MRR, 0.5965 to 0.4843, is 0.1122 and is larger than the sampling interval, whereas the difference between adjacent low weights is not. Exact skill matching helps only as a low-weight tiebreaker, because resume skill text and taxonomy labels draw on different vocabularies, so exact overlap fires rarely and often on generic terms.

8.2 Resume Parsing — prompt experiments

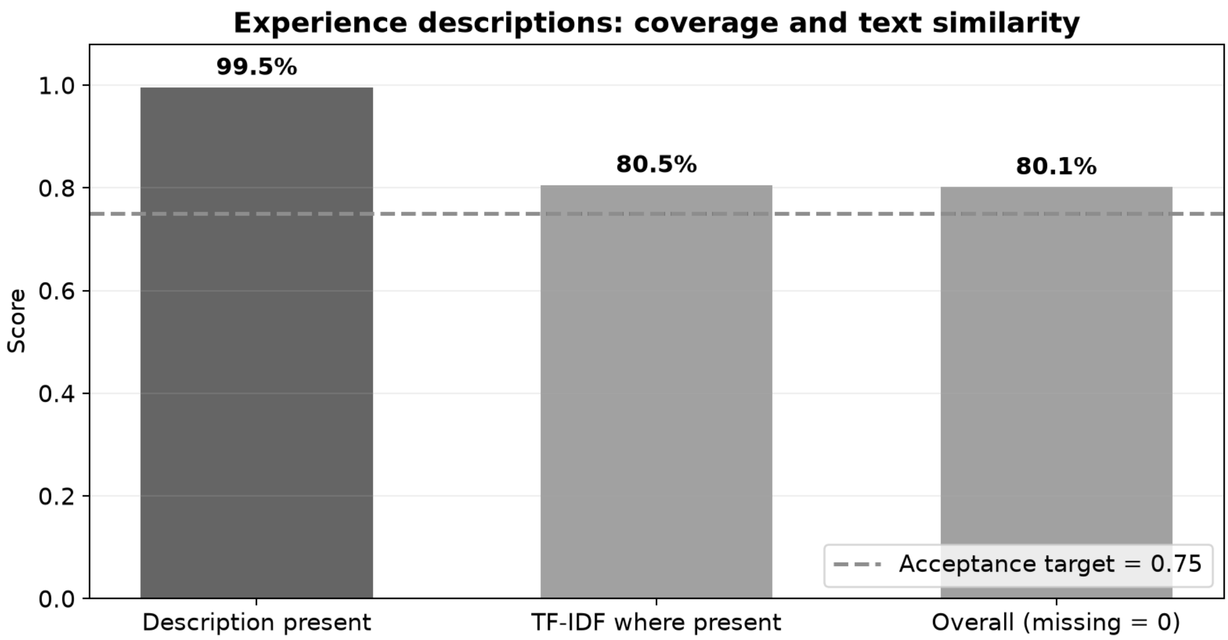


Figure 5. Diagnostic macro-average of section-level normalised micro-F1 across the prompt experiments. The step between the Gemma and Gemini series is larger than anything prompt wording achieved within either series.

Experiment	Measured result	Decision
Broad skill search across the whole document	Skills F1 0.2268 → 0.1830; false positives 34 → 87	Rejected — duties were converted into skills

Atomic-skill splitting examples	Skills F1 0.6099 → 0.6115; precision 0.5811 → 0.5333	Rejected — negligible F1 gain, lower precision
Source-faithful boundaries	Skills F1 0.8198; Projects F1 0.8814	Accepted
Strict complete-line skills	Skills F1 0.8198 → 0.5607	Rejected — valid reference entities were omitted
Visual project boundaries	Projects F1 0.1724 → 0.4550	Retained — boundaries improved; named works still missed
Strict credential boundaries	Certifications F1 0.7472 → 0.6503	Rejected — global extraction loss exceeded the boundary correction

Search scope mattered more than asking for completeness, and the two failure modes are symmetric: splitting grouped phrases raised recall and lowered precision, while preserving every printed line intact omitted tool embedded inside those lines. Stable aliases and unambiguous enumerations were therefore moved into deterministic comparison-time normalisation, and uncertain omissions and inventions were retained as model errors.

8.3 Rejected configurations

Rejected	Measured outcome	Reason
Skill-bonus weights 0.05–0.40	MRR falls from 0.5866 to 0.4843	Monotonic degradation; overlap is a tiebreaker, not a ranking signal
Previously specified re-ranking rule	MRR 0.4949, Hit@5 0.6731 held-out	Sits below the retrieval ordering it was meant to improve
Five alternative ranking weight sets	Composite 0.7503–0.7670 against 0.8977	Latency above 4,000 ms with no compensating quality gain
Four alternative search and crawl sets	Crawl 0.067–0.45; latency 6,868–26,604 ms	Lower crawl success and higher latency than narrow_fast
Five alternative judge prompts	Schema 0.0000–0.5000; pass rate 0%	None passes; the retained variant is least unreliable, not acceptable
Hosted embedding model for the occupation index	Build failed at 100 of 3,039 documents	Free-tier quota of roughly 100 requests per minute and 1,000 per day

8.4 Justification of the composite weighting

The latency term carries 15 percent of the composite. That value was not derived mathematically; it was set so latency could break a tie between configurations of equivalent quality without overturning a real quality difference. What can be shown is that the reported selections do not depend on it.

Sweep	Sensitivity of the selection to the latency weight	Consequence
Job Matching ranking	Quality metrics vary by at most 0.017 across the whole space while latency varies eighteen-fold; balanced_50_50 wins at any non-zero latency weight and ties on quality at zero	Selection is invariant; the weight is not load-bearing
Job Discovery search and crawl	narrow_fast is simultaneously highest on crawl success and lowest on latency	Selection is invariant; it wins on a dominance argument, no weighting needed
Prompt sweeps	Latency enters as a hard pass/fail ceiling rather than a weighted term	No weighting sensitivity; the judge fails the ceiling outright
Career Recommendation	No latency term; selected on MRR and Hit Rate alone	Not applicable

In every sweep the selected configuration is either dominant on all dimensions or invariant to the weight across its plausible range, so the composite is a reporting convenience rather than the basis of any decision. The weight remains an unjustified constant and is labelled heuristic; a principled alternative would be to state a latency budget as a constraint and maximise quality subject to it, which would remove the free parameter entirely.

9. Ablation Study

Component	Measure	With	Without	Contribution
Gemini 3.5 Flash parser	Skills F1	0.82	0.53	+0.29
Gemini 3.5 Flash parser	Projects F1	0.88	0.45	+0.43
Semantic similarity as primary signal	MRR, 86 resumes	0.5965	0.4843	+0.1122, separable
Cosine distance function	Retrieval Hit@5, 86 resumes	0.8256	0.7791	+0.0465, not separable
Deterministic re-ranking step	Hit@5, held-out 52	0.8269	0.8077	+0.0192, not separable
Skill-overlap bonus at weight 0.02	MRR, 86 resumes	0.5965	0.5849	+0.0116, not separable
Lexical signal in Job Matching	Precision@5	0.8486	0.8419	+0.0067, within noise
Semantic signal in Job Matching	Precision@5	0.8486	0.8512	−0.0026, within noise

Three readings follow. The parser change is the largest measured effect anywhere in the system and is separable in every section. Within Career Recommendation the semantic signal is doing the work — driving the skill weight to 0.40 approximates removing similarity as the primary signal and costs 0.1122 on MRR. Everything else in that module, including the entire deterministic re-ranking stage, produces differences smaller than the sampling interval and is reported as not separable rather than as a contribution.

The Job Matching ablation is a negative result: neither signal can be shown necessary on this pool. The hybrid score is retained on reasoning grounds — the lexical component catches exact tool and certification names that embeddings blur, the semantic component catches equivalent skills phrased differently — but that reasoning is not currently supported by measurement, and a ten-job pool is very likely too small to test it. The judge stage cannot be ablated cleanly, because it contributes 0.2 of the final score but does not yet produce reliably parseable output. For Job Discovery, removing the breadth constraint drops crawl success from 0.50 to 0.067 and raises latency six-fold with coverage unchanged.

10. Error Analysis

10.1 Career Recommendation: quantified failure decomposition

Because Hit@20 is measured on the same profiles as Hit@5, the misses decompose exactly rather than by estimate. 13 of 86 profiles had no acceptable occupation in the top seven.

Cause	Count	Share of misses	Evidence
Truncation: present in top 20, lost at top 7	6	46%	Hit@20 0.9186 against Hit@7 0.8488; the re-ranker diagnostic records 6 dropped out of top 7
Never retrieved: absent from top 20	7	54%	86 – 79 = 7
of which taxonomy coverage gap	≈6	46%	Arts, Management and Network Security Engineer: 2 records each, all missed
of which other retrieval failure	≈1	8%	Remainder

Truncation is the smaller half of the error and is already partly bought back. Moving the final list from five to seven raises Hit@7 to 0.8488, 73 of 86, against Hit@5 of 0.8372, 72 of 86 — one profile recovered for two extra slots. Six acceptable occupations still sit between ranks 8 and 20, so the remaining truncation loss is distributed rather than concentrated, and widening the list further buys progressively less.

Vocabulary mismatch is deliberately not listed as a separate row with a percentage, and this is a substantive point rather than an omission. It is not separable at profile level: it degrades the skill-overlap signal across every profile rather than causing specific identifiable misses. Its measured effect is the monotonic decline of MRR from 0.5965 to 0.4843 as the skill weight rises from 0.02 to 0.40. Assigning it a share of the 13 misses would double-count profiles already attributed to truncation or coverage, so it is quantified by its effect size instead. Encoder truncation is likewise unquantified at profile level: the local encoder caps at 512 tokens against occupation documents reaching roughly 1,300, so the tails of the longest documents are never embedded, but the affected subset has not been isolated, and no per-profile attribution is claimed.

10.2 Resume Parsing: confusion examples

Each error class below was diagnosed from side-by-side comparison with the source document, with the resolution recorded.

Error class	Prediction against reference	Diagnosis and resolution
Contact: casing	TIMOTHY DUNCAN against Timothy Duncan	Surface variant; resolved by case-insensitive comparison
Contact: over-extraction	Full street address against locality only	Unnecessary personal data; resolved by storing and comparing locality only
Contact: genuine error	Tara Walton against Janice Walton	Incorrect extraction; retained as FP and FN, normalisation must not hide it
Education punctuation	PhD against Ph.D.	Variant; resolved by controlled degree aliases
Education: boundary	Institution returned with trailing locality	Field-boundary error; resolved by removing verified trailing locality
Experience: date pairing	“2016 - present” in start_date	Range placed in one field; resolved by splitting visible ranges into paired endpoints
Experience: OCR	Employer name misread against the printed name	Genuine content error; retained, no fuzzy matching in the official score
Skills: false positive	A grouped label retained instead of its named tools	Source-boundary error; retained as FP
Skills: false negative	Grouped source line against individually annotated tools	Controlled aliases and enumeration splitting recover safe cases; unresolved items remain FN
Projects: false positive	Project title contaminated by its description text	Correct project, incorrect name boundary
Projects: false negative	Nine visibly named patents returned as no projects	Genuine extraction omission of a whole class of named work
Certifications: false positive	“AWS Cloud Practitioner Validation Number 246DCT7D1BIQRS” against “AWS Cloud Practitioner”	Identifier retained inside the credential name; resolved by deterministic identifier cleanup
Certifications: false negative	A visibly completed training entry omitted	Genuine extraction omission

The boundary between normalisation and error is held deliberately: missing, hallucinated or materially incorrect values remain FP or FN, OCR differences in names and employers are not fuzzy matched in the official score, and technical punctuation stays meaningful, so C, C++, C# and .NET are not interchangeable. Skills text similarity is reported beside the entity score as a diagnostic only, its mean of 0.53 does not replace the item-level F1 of 0.82, because unrelated skill lists can share common words.

10.3 Job Discovery and Matching

Discovery-stage failures are source-side rather than pipeline defects: a representative run found five unique URLs, attempted three, extracted none, and completed cleanly by skipping the failed URLs. Postings that render client-side, block automated access, or omit structured metadata leave the zero-model extraction stage nothing to work from, and the sweep confirms this because success falls as breadth rises while coverage stays at 1.0. The judge stage fails in two distinct ways under the same prompt at temperature 0 — prepending visible reasoning before the JSON array, which breaks the parser, and exceeding the 20-second ceiling, with one case reaching 178,664 ms. Both are consistent with the token-overflow root cause in Section 6.3.3.

10.4 Defects found during integration

The most consequential defect produced no error at any layer. During the migration, the vector-store client's insertion helper wrote malformed embeddings: content, metadata and row counts were all correct, so it passed every surface check. It was detectable only by measuring the vectors themselves — a stored vector had an L2 norm of 0.5951 where this encoder always produces approximately 1.0, and comparing a stored vector against a fresh embedding of its own document text gave a cosine similarity of −0.0072, so a document was orthogonal to itself. The user-visible symptom was retrieval nonsense. The fix bypassed the helper to write the vector literal directly through the database driver, and a verification step now ships inside the ingestion script, so a rebuilt index is checked for vector norm and self-similarity rather than trusted because the insert succeeded.

Defect	Symptom	Root cause and resolution
Parser schema-adherence failure	Valid JSON, but 0 of 5 smoke-test responses matched the production schema — work_history for experience, employer for company, graduation_date for education.end_year	The integration treated system-role support as evidence of schema-constrained decoding. Resolved by routing through the structured-response interface, parsing against the Pydantic model, and inserting the full JSON Schema into the prompt as a second safeguard. The same cohort moved to 5 of 5
Search-function signature mismatch	Every retrieval call returned PGRST202	The client sent only the query embedding plus an optional filter and applied the limit client-side, while the SQL function required an explicit count argument; function recreated to match the caller's contract
Silent row-limit cap	A check against the 3,039-row table returned 1,000 rows	REST-layer default row limit; the check would have concluded the table was under-populated and re-ingested roughly 2,000 duplicates. Caught before the re-run; the check now paginates
Vendor model retirement, three occurrences	A configured identifier 404s for new keys; a second is unavailable; a hosted embedding endpoint is quota-blocked	Provider-side deprecation and quota policy; mitigated by holding every model identifier in environment configuration

11. Robustness

11.1 Generalisation to unseen data

The held-out comparison is the primary generalisation evidence: the configuration selected on the annotated set scored at or above its full-set figures on every measure when run on the 52 unseen resumes. For Job Matching the equivalent evidence is stability — ranking quality holds across six materially different weight configurations with none collapsing. Three limits are visible in the evidence: the resume corpus is entirely single-page and image-only, so the text path and multi-page merge are less exercised than the vision path; the occupation taxonomy is European in origin, so regional and emerging titles map less cleanly, which is directly visible in the network-security category; and the Job Matching pool is small and curated.

11.2 Fail-soft behaviour

Condition	Behaviour
Parser output fails schema validation or the completeness check	One fallback retries against a second model, then a partial profile with review flags rather than nothing
Candidate skills produce no overlap with any retrieved occupation	Similarity-only ranking, confidence reported as low, with an explicit message to the user
Explanation model unavailable, or no candidates survive retrieval	Explicit degraded or no-candidates status with the deterministic ranking intact
Explanation model names an occupation outside the retrieved set	Entry dropped by URI validation before the response is returned
Primary job API returns nothing or fails	Falls back to the SearXNG and crawl chain; an empty shortlist carries a no-matches status rather than raising

Across the annotated evaluation the result of this design is that 86 of 86 profiles produced a ranked list and none failed.

11.3 Stress testing: what was tested and what was not

The table separates the stress scenarios that have been exercised from those that have not.

Scenario	Status	Evidence or reason
Scanned documents of varying quality	Tested	A robustness set was generated from the corpus using small rotations, moderate Gaussian blur, JPEG compression, contrast and resolution reduction, and light image noise, each augmented document inheriting the annotation and split of its source
Missing sections	Tested	The parser returns null for missing scalars and empty lists for missing sections; the integration run confirmed sections absent from the source returned empty rather than populated by inference. Projects appear in only 5 gold records and certifications in 27, so the absent case is well represented
Adversarial instruction text inside a resume	Tested by construction	The extraction prompt treats the document as data and not instructions, and the output schema is a hard gate; no formal adversarial suite was executed, so this is an implemented control rather than a measured defence
Load and concurrency	Open	No load testing was performed

The three open rows share one cause: the evaluation corpus is single-page, English, and image-only, so scenarios outside that envelope cannot be measured against it without sourcing new data. Claims about production readiness are therefore restricted to that envelope.

12. Computational Performance

12.1 Latency

Stage	Measured	Notes
Resume parsing, per resume	19.03 s median, 21.79 s mean	All 86 resumes on the selected model; maximum 80.26 s. Gemma 4 baseline: 17.21 s median, 126.51 s maximum, 83 of 86 completed
Career Recommendation, per profile	1.5 to 1.8 s	132–158 s across 86 profiles against the hosted store
Job Matching ranking	226 ms	Against more than 4,000 ms for every rejected configuration
Job Discovery, API path	Within 2 minutes for successful queries	Across 45 domains retrieved

Job Discovery, fallback crawl	2,703 ms	Against 26,604 ms for the slowest configuration tested
Query generation	19 ms	Range 19 to 33 ms across seven variants
Batched judge	21,572 ms median	Exceeds the 20-second ceiling; one case reached 178,664 ms
Occupation index build	Minutes on CPU	One offline pass over 3,039 documents with the local encoder

Centralising the vector store raised Career Recommendation evaluation time from 11.7 seconds to 132–158 seconds for the same 86 profiles. The cause is network round-trip time to the hosted region rather than additional computation, and it is a per-query cost rather than a scaling problem, since the index is queried once per profile regardless of its size. The batched judge exceeding its own ceiling at the median is an open defect, not an acceptable measurement.

12.2 Resource footprint and cost

Resource	Footprint
GPU	Zero, everywhere, by design; the only local model is a 110M-parameter CPU encoder
Memory	Approximately 600 MB to 1 GB for the application stack, plus 100 to 200 MB per concurrent headless browser context in the crawler
Stored vectors	3,039 vectors of 768 dimensions, roughly 9 MB of raw float data plus index structure and document text
Database storage	Roughly 0.3 to 0.5 GB available on the free tier; the job store is deduplicated by a hash of the normalised posting URL, so storage grows with distinct postings rather than with searches
Call budget	One call per resume page plus at most one fallback; one local embedding and one explanation call per recommendation; exactly two model calls per Job Discovery run regardless of job count
Token accounting	Per-call token counts are recorded in the append-only evidence ledger for every parser run, so a monetary figure is derivable from published rates

The binding cost constraint was never money but free-tier quota, which is what forced the local encoder, the offline index build, the batched judge, and the deterministic extraction stage. Tokens are recorded per call, but no monetary cost per resume is reported, and no throughput, concurrency or scaling measurement was performed, the call-budget row describes design intent rather than observed behaviour under load. The single-user latency figures should not be read as throughput estimates, because the crawler's per-context memory cost and the hosted providers' rate limits are both expected to bind before CPU does.

13. Limitations

13.1 Dataset and methodology

- 58 of the 86 gold records have been source-verified; the remaining 28 retain AI-drafted annotations, so a residual accuracy ceiling applies to every metric computed over the full set.
- The Career Recommendation ground truth is a keyword-generated, hand-reviewed proxy rather than direct occupation labels, and several categories map to overlapping occupation sets.
- At n = 86 the 95 percent interval on a rate near 0.83 is roughly plus or minus 0.08, so several reported differences — including the re-ranker's contribution — are not statistically separable, and are described as such throughout.
- The live-search evaluation improved in domain coverage but remains approximately one query per domain, so per-domain depth is shallow and the reported reliability figures are point estimates on very small samples.
- The skill-bonus weight sweep was run over all 86 records including the 52 held out, so for that one parameter selection and confirmation are not strictly separated; the held-out comparison for the selected value was run separately and is clean.

- No human evaluation of recommendation usefulness, explanation quality, or user satisfaction was conducted. This is the largest remaining methodological gap for a system whose output is advisory.

13.2 Model and system

- The batched judge has no passing configuration across six variants. The root cause is identified as token overflow and a staged remedy is specified, but none of it is built, so the stage remains an unresolved reliability gap rather than a working component.
- Crawl success remains 0.50 on the best configuration. The primary API layer reduces how often the crawl path is reached but does not repair it, and the API does not serve niche domains — precisely the case that falls through to the crawl.
- The reported crawl-success figure measures the crawl layer in isolation; the two-layer pipeline has not been swept end to end, so no system-level discovery reliability number is claimed.
- The local encoder truncates at 512 tokens against occupation documents reaching roughly 1,300, so the longest documents are not embedded in full, and the affected subset has not been isolated.
- Several values remain heuristic: shortlist size, the hybrid and judge weights, the essential and optional skill weights, and the 15 percent latency term in the composite.

13.3 Engineering and operational

- Two deployment-hygiene items remain open: route prefixes are inconsistent across modules, and the central configuration module still carries retired model identifiers as default values, so a deployment without environment overrides would start against models that no longer exist.
- Development API credentials circulated through team debugging sessions during this milestone. They are development-scope keys held in environment configuration, not production credentials, and are treated as requiring rotation before any shared deployment.

13.4 Bias, privacy, and ethical evaluation

Concern	Status	Evidence or gap
Privacy	Verified	Email addresses and telephone numbers are never extracted or persisted, enforced at the prompt and schema level and confirmed on the integration run; locations retain locality only; original uploads and rendered page images are not written to persistent storage; every profile access is written to a separate audit log; retained sensitive fields are encrypted at field level and decrypted only for the authenticated owner
Explainability	Partly verified	Every occupation URI returned by the explanation model is validated against the retrieved set and a non-matching entry is dropped, so explanations cannot cite occupations the system did not retrieve. Whether the explanations are useful to a reader is not measured, that requires the human study in Section 4.6
Regional bias	Measured indirectly	A European occupational taxonomy carries the labour-market structure of the region that produced it. This is visible in the per-category results rather than hypothetical: Network Security Engineer has no counterpart in the taxonomy and all its records miss
Corpus composition bias	Measured	The source corpus over-represents some categories by a factor of roughly twelve; the gold set counters this by sampling two resumes per category
Gender bias	Open	Not measured. The gold set carries no gender annotation, and no counterfactual test, re-running matched profiles with names and gendered terms substituted, was constructed. The parser excludes contact identifiers, but names remain in the extracted profile and reach the embedding, so a bias channel exists and is untested
Decision-making scope	Stated	The match score is a deterministic decision-support value and explicitly not a probability, and the judge's interview probability is not calibrated against real hiring data. Presenting either as a prediction of

		employment outcome would be unsupported by anything measured here

14. Possible Improvements

Each option follows from a specific measurement in this report and is stated with its cost rather than as a commitment. They are ordered by the size of the measured problem they address.

Improvement	Follows from	Cost
Run the human evaluation protocol in Section 4.6	No usefulness, explanation-quality or satisfaction evidence exists	Two raters over 20–30 profiles, plus agreement scoring before any figure is reported
Build the judge token-budgeting and structured-extraction fix	Six variants at a 0 percent pass rate with token overflow as root cause	Pre-flight token counting plus a field-extraction step; removes most of the payload without a second model call
Sweep the two-layer discovery pipeline end to end	The 0.50 figure measures the crawl layer alone	Re-run the existing sweep methodology over the API-plus-fallback chain
Complete gold verification for the remaining 28 records	58 of 86 verified; the rest cap metric accuracy	Side-by-side source review, the same process already applied to 58
Construct a gender counterfactual test set	A bias channel exists through names in the embedded profile and is untested	Matched profile pairs with substituted names and gendered terms; measures rank shift on identical qualifications
Close the vocabulary gap	MRR falls 0.5965 to 0.4843 as exact matching gains weight	Embedding the 13,939 skill labels for semantic matching, a curated alias table, or a cross-encoder re-ranker over the top twenty, in ascending order of cost
Measure encoder truncation in isolation	512-token cap against ~1,300-token documents, effect not isolated	Score the affected occupation subset before changing anything
Add throughput and cost instrumentation	Tokens are recorded but no cost, throughput or concurrency figure exists	Derive cost per resume from the existing ledger; add a concurrent load harness
Extend the corpus beyond single-page English	Multilingual and long-resume behaviour is unknown, not merely unmeasured	Requires sourcing new annotated data, the largest single cost on this list

15. Conclusion

All three modules now run inside a single application over a centralised database, which completes the architecture proposed at Milestone 3. Resume Parsing has a complete field-level evaluation over all 86 gold resumes, with every section meeting the 0.75 threshold and the provider selection measured against an explicit baseline. Career Recommendation carries confidence intervals and a significance test, and its failure decomposition is exact rather than qualitative.

Two of those results are negative and are reported as such. The re-ranking step is not shown to improve Hit@5 — the difference is one profile at p = 1.00 — and the earlier claim of a lift is withdrawn. The truncation remedy is worth far less than the failure decomposition alone suggested: two extra list slots recover one profile, not seven.

Component	Headline result	Status
Resume Parsing	All six sections ≥ 0.75 F1; Contact 0.98, Experience 0.98, Education 0.92, Projects 0.88, Skills 0.82, Certifications 0.75	Measured, passing
Career Recommendation	MRR **0.6121**, Hit@1 0.4884, Hit@5 0.8372, Hit@7 0.8488, Hit@20 0.9186, 0 failures in 86	Measured; re-ranker gain not significant

Career Recommendation	MRR 0.6359, Hit@5 0.8269 on 52 unseen resumes	Measured, generalises
Job Matching ranking	Precision@5 0.8486, NDCG@10 0.9795, Spearman 0.8767 at 226 ms	Measured; efficiency result, quality flat
Job Discovery	Coverage 1.0; API primary layer validated across 45 domains; fallback crawl 0.50	Measured; reliability below acceptance
Query generation	Schema 1.0000, diversity 0.8010, pass rate 100%	Measured, passing
Batched judge	Schema 0.5000, calibration 0.5000, pass rate 0%	Measured, failing; root cause identified
Integration	All modules over one application and one database	Complete

The Resume Parsing and Career Recommendation path is ready to be used as decision support and is not ready to be used as an automated decision, which matches how it is presented to the user: a short, ranked list with confidence bands and an explanation grounded in retrieved evidence. Job Discovery is now architecturally complete and functionally incomplete — its judge stage has a diagnosed but unbuilt fix, and its crawl fallback remains unreliable.

Three gaps are carried forward without mitigation: no human evaluation, no gender-bias measurement, and no testing outside a single-page English image-only envelope. Each is a real limit on what this evaluation can support, and each is stated rather than absorbed into a favourable summary. The state of the system is that its measured components are measured properly and at stated confidence, its unmeasured components are named as unmeasured, and its failing component is reported as failing.

16. Team Consent

Name	Roll Number	Initial
Gaurav Kumar	22f1001105	GK
Dev Gupta	22f2000888	DG
Abhinav Ohri	24f1002064	AO
Pranav N	22f2000117	PN
Praveena N	22f3001454	PN